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STATISTICAL MECHANICS OF TWO-DIMENSIONAL COULOMB SYSTEMS

I. LATTICE SUMS AND SIMULATION METHODOLOGY

John W. PERRAM

Matematisk Institut, Odense University, Campusvej 55, 5230 Odense M, Denmark

and

Simon W. de LEEUW

Theoretical Physics Department, 1 Keble Rd., Oxford, OX1 3NP, England

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The statistical mechanics of systems acting via two-dimensional charge–charge and dipole–dipole interactions is studied in periodic boundary conditions. The two-dimensional lattice theta-function transformation is obtained and the sum shown to contain a polarization correction to the Ewald sum, analogous to that found in three dimensions. Using a different approach, these sums are evaluated in closed form, for bodies of arbitrary shape. The methodology for the computer simulation of two dimensional dielectrics is discussed and the two-dimensional analogue of a generalized Kirkwood–Clausius–Mosotti–Debye formula derived.

1. Introduction

With the application of lattice sum techniques to the computer simulation of systems with Coulomb interactions, a number of things have recently become clear. Firstly, the well-known Ewald sum for perfect crystals is not the correct value of the lattice sum of unit cells containing an arbitrary (but neutral) distribution of charges,[1,2,3]). It has further been shown that such sums depend on the surroundings of the body[2]), and claimed[4]) and shown[5]) that the sums depend on the shape of the large volume containing the periodic charge or dipole distribution.

For a set of N charges $q_1, q_2, \dots, q_N$ at positions $r_1, r_2, \dots, r_N$ within a cubic unit cell of side L, the lattice sum over an infinite spherical assembly S_0 of such cubes is

$$U^c(N; S_0) = \tfrac{1}{2} \sum_{L\boldsymbol{n}\in S_0}{}' \sum_{\alpha=1}^{N} \sum_{\beta=1}^{N} q_\alpha q_\beta |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta + \boldsymbol{n}L|^{-1}, \tag{1.1}$$

where the sum is over all lattice vectors $\boldsymbol{n}L$ within S_0, the prime indicating that the term $\boldsymbol{n} = \boldsymbol{0}$ is to be omitted if $\alpha = \beta$. We have[2])

$$U^{c}(N; S_0) = U(N; E) + 2\pi/(3L^3)\Big(\sum_{\alpha=1}^{N} q_\alpha \boldsymbol{r}_\alpha\Big)^2, \tag{1.2}$$

where $U(N; E)$ is the usual Ewald sum, which may also be regarded as the sum over a lattice on the surface of a 4-dimensional hypertorus. It has been further shown[2]) that the lattice sum $U(N; S_0, \epsilon')$ where the sphere S_0 is surrounded by a spherical shell of uniform dielectric material with dielectric constant ϵ' is

$$U(N; S_0, \epsilon') = U(N; E) + 2\pi L^{-3}(2\epsilon' + 1)^{-1}\Big(\sum_{\alpha=1}^{N} q_\alpha \boldsymbol{r}_\alpha\Big)^2, \tag{1.3}$$

so that the Ewald sum corresponds to a spherical array surrounded by a conductor. This accounts for the puzzling observation that computer simulations of molten salts using the Ewald sum give a non-zero conductivity. The term additional to the Ewald sum, which is proportional to the square of the cell dipole moment, plays a decisive role in determining the simulated values of dielectric properties[2,7,8]), and conductivity[6]).

Recently, simulations have been reported[9,10]) for two-dimensional dipolar[9]) and charged[10]) systems, where the Coulomb potential is

$$\phi^{c}_{\alpha\beta} = -q_\alpha q_\beta \log(|\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|), \tag{1.4}$$

without awareness of the discussion of the role of boundary conditions for three-dimensional systems. It is the purpose of this article to set out that discussion for two-dimensional systems. In section 2, we derive the two dimensional analogue of the Ewald transformation for square cells, showing that the lattice sum for circular regions D_0, is, like eq. (1.2), not periodic. In section 3 we shall, by writing

$$z_\alpha = x_\alpha + \mathrm{i}y_\alpha, \tag{1.5}$$

where $\boldsymbol{r}_\alpha = (x_\alpha, y_\alpha)$, obtain the result that

$$\begin{aligned} U^{c}(N; D_0, 1) &= -\tfrac{1}{2}\sum_{ln\in D_0}{}' \sum_{\alpha=1}^{N}\sum_{\beta=1}^{N} q_\alpha q_\beta \operatorname{Re}\log(z_\alpha - z_\beta + 2\Omega_{m,n}) \\ &= -\tfrac{1}{2}\operatorname{Re}\sum_{\alpha=1}^{N}\sum_{\beta=1}^{N} q_\alpha q_\beta \log\Big\{\frac{\sigma(z_\alpha - z_\beta)}{z_\alpha - z_\beta}\Big\} \\ &\quad -\tfrac{1}{2}\operatorname{Re}\sum_{\alpha=1}^{N}\sum_{\beta\neq a} q_\alpha q_\beta \log(z_\alpha - z_\beta), \end{aligned} \tag{1.6}$$

where $\sigma(z)$ is the Weierstrass sigma function, and $\Omega_{m,n} = (l/2)(m + \mathrm{i}n)$, where $\boldsymbol{n} = (m, n)$ and l is the side of the square. The corresponding result for dipoles is obtained. The lattice sum $U^{c}(N; D, 1)$, over any simply connected region D

bounded by a simple closed curve Γ, is shown to be

$$U^{c}(N;D,1)=U^{c}(N;D_0,1)+\mathrm{Re}(\mathrm{i}/(4l^2)\Big(\sum_{\alpha=1}^{N}q_\alpha z_\alpha\Big)^2\oint_{\Gamma}\overline{\mathrm{d}z}/z \tag{1.7}$$

where $\bar{z}=x-\mathrm{i}y$ is the complex conjugate of z.

In section 4, we study fluctuations of the polarization $\boldsymbol{M}=\Sigma_{\alpha=1}^{N}\boldsymbol{\mu}_\alpha$, where $\boldsymbol{\mu}_\alpha$ is the dipole moment, and derive the formula

$$\frac{(\epsilon-1)(\epsilon'+1)}{2(\epsilon+\epsilon')}=\left(\frac{\pi\rho\mu^2}{2k_{\mathrm{B}}T}\right)\left\{\frac{\langle M^2\rangle_{\epsilon'}-\langle M\rangle_{\epsilon'}^2}{N\mu^2}\right\} \tag{1.8}$$

for the dielectric constant ϵ when the ensemble averages $\langle \boldsymbol{M}^2\rangle_{\epsilon'}$, $\langle \boldsymbol{M}\rangle_{\epsilon'}$ are computed for a system acting with the dipole Hamiltonian using the dipole analogue $U^{d}(N;\ S_0,\ \epsilon')$ of eq. (1.3). Here $\rho=Nl^{-2}$ is the number density, $\mu=|\boldsymbol{\mu}_\alpha|$ the dipole moment, k_{B} Boltzmann's constant and T the absolute temperature. By setting $\epsilon'=1$, ϵ and ∞ in eq. (1.8) we obtain the two-dimensional analogues of the Clausius–Mosotti, Kirkwood and Debye formulae.

2. The two-dimensional theta-function transformation

We seek to compute

$$U^{c}(N;D_0,1)=-\tfrac{1}{2}\sum_{l\boldsymbol{n}\in D_0}{}'\sum_{\alpha=1}^{N}\sum_{\beta=1}^{N}q_\alpha q_\beta\log(|\boldsymbol{r}_\alpha-\boldsymbol{r}_\beta+\boldsymbol{n}l|) \tag{2.1}$$

as economically as possible for square cells of side l lying within a large circle D_0, by splitting it into two rapidly convergent sums. We note that, as $|\boldsymbol{n}|\to\infty$, and provided $\Sigma_{\alpha=1}^{N}q_\alpha=0$,

$$\sum_{\alpha=1}^{N}\sum_{\beta=1}^{N}q_\alpha q_\beta\log(|\boldsymbol{r}_\alpha-\boldsymbol{r}_\beta+\boldsymbol{n}|)\sim|\boldsymbol{n}|^{-2} \tag{2.2}$$

in general, so that the series is conditionally convergent. We study the series

$$U^{c}(N;D_0,1,s)=-\tfrac{1}{2}\sum_{l\boldsymbol{n}\in D_0}{}'\mathrm{e}^{-s\boldsymbol{n}^2}\sum_{\alpha=1}^{N}\sum_{\beta=1}^{N}q_\alpha q_\beta\log(|\boldsymbol{r}_\alpha-\boldsymbol{r}_\beta+\boldsymbol{n}l|), \tag{2.3}$$

which is absolutely and uniformly convergent in s for $\mathrm{Re}(s)>0$ and which adds up the sum in circular annuli, since it applies the same convergence factor $\exp(-s\boldsymbol{n}^2)$ to all terms with the same $|\boldsymbol{n}|$. By studying the integral

$$-\frac{1}{2\pi\mathrm{i}}\int_C\left\{\sum_{j=1}^{N}q_j\log(\rho+\gamma_j^2)\right\}\exp(\rho t)\,\mathrm{d}\rho \tag{2.4}$$

around a large contour C consisting of the imaginary ρ-axis from $-\infty$i to ∞i, a large semi-circular contour in the left-half plane and the negative real axis from $-\infty$ to 0^- and from 0^- to $-\infty$, indented at the points $-\gamma_j^2$, we obtain the result

$$\int_0^\infty \sum_{j=1}^N q_j \exp(-\gamma_j^2 t)t^{-1}\,\mathrm{d}t = -\sum_{j=1}^N q_j \log \gamma_j^2 \tag{2.5}$$

valid for $\mathrm{Re}(\gamma_j^2) > 0$. Thus

$$U^c(N; D_0, 1, s) = \tfrac{1}{4}\sum_{n l \in D_0} \mathrm{e}^{-sn^2} \int_0^\infty \sum_{j=1}^N \sum_{k=1}^N q_j q_k t^{-1} \exp(-t(\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n})^2)\,\mathrm{d}t, \tag{2.6}$$

where $\boldsymbol{R}_j = \boldsymbol{r}_j/l$.

In the usual way, we write

$$\int_0^\infty \mathrm{d}t = \int_0^{\alpha^2} \mathrm{d}t + \int_{\alpha^2}^\infty \mathrm{d}t$$

and obtain

$$U^c(N; D_0, 1, s) = U_{\mathrm{I}}(s) + U_{\mathrm{II}}(s) \tag{2.7}$$

by splitting the integral. The parameter α^2 is chosen to give optimum convergence. The sum $U_{\mathrm{II}}(s)$, involving $\int_{\alpha^2}^\infty \mathrm{d}t$ is absolutely convergent for $s = 0$, and we have that

$$U_{II}(s) = \tfrac{1}{4}\sum_n \sum_{j=1}^N \sum_{k=1}^N q_j q_k E_1(\alpha^2 |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^2) - \tfrac{1}{4}(\gamma + \log \alpha^2)\sum_{j=1}^N q_j^2. \tag{2.8}$$

Here $E_1(z)$ is the exponential integral function and γ is Euler's constant.

To evaluate $U_{\mathrm{I}}(s)$, which contains the divergence at $s = 0$, we write

$$-s\boldsymbol{n}^2 - t(\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n})^2 = -t(\boldsymbol{R}_j - \boldsymbol{R}_k)^2 - 2\boldsymbol{n}\cdot(\boldsymbol{R}_j - \boldsymbol{R}_k)t - (s+t)\boldsymbol{n}^2 \tag{2.9}$$

and we use the standard result[11])

$$\begin{aligned}\sum_{\boldsymbol{n}} \exp[-2\boldsymbol{n}(\boldsymbol{R}_j - \boldsymbol{R}_k) - (s+t)\boldsymbol{n}^2] &= \theta_3\left(\mathrm{i}(X_j - X_k)t \,\middle|\, \frac{\mathrm{i}(s+t)}{\pi}\right) \\ &\quad \times \theta_3\left(\mathrm{i}(Y_j - Y_k)t \,\middle|\, \frac{\mathrm{i}(s+t)}{\pi}\right) \\ &= \frac{\pi}{(s+t)} \exp(t^2(s+t)^{-1}(\boldsymbol{R}_j - \boldsymbol{R}_k)^2) \\ &\quad \times \theta_3\left(\frac{\pi t(X_j - X_j)}{s+t} \,\middle|\, \frac{\pi}{(s+t)}\right) \\ &\quad \times \theta_3\left(\frac{\pi t(Y_j - Y_k)}{(s+t)} \,\middle|\, \frac{\pi}{s+t}\right),\end{aligned} \tag{2.10}$$

which, in the notation of[11]) is the imaginary transformation of the Jacobi theta function $\theta_3(z \mid \tau)$. Performing manipulations similar to those in ref. 2, and making an asymptotic expansion in powers of s^{-1} as $s \to 0$, we find

$$U_1(s) = \frac{1}{4\pi} \sum_{n \neq 0} |\boldsymbol{n}|^{-2} \exp(-\pi^2 \boldsymbol{n}^2 \alpha^{-2}) \left| \sum_{j=1}^{N} q_j \exp(2\pi i \boldsymbol{n} \cdot \boldsymbol{R}_j) \right|^2 - (\pi/4) \sum_{j=1}^{N} \sum_{k=1}^{N} q_j q_k (\boldsymbol{R}_j - \boldsymbol{R}_k)^2 + \mathcal{O}(s). \tag{2.11}$$

Combining eqs. (2.7), (2.8) and (2.11), in which the limit $s \to 0$ may now be taken, we obtain the two-dimensional lattice sum

$$\begin{aligned} U^{c}(N; D_0, 1) = &-\tfrac{1}{4}(\gamma + \log \alpha^2) \sum_{j=1}^{N} q_j^2 + \tfrac{1}{4} \sum_{\boldsymbol{n}}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} q_j q_k E_1(\alpha^2 |\boldsymbol{R}_j - \boldsymbol{R}_k|^2) \\ &+ \frac{1}{4\pi} \sum_{n \neq 0} |\boldsymbol{n}|^{-2} \exp(-\pi^2 \boldsymbol{n}^2 \alpha^{-2}) \left| \sum_{j=1}^{N} q_j \exp(2\pi i \boldsymbol{n} \cdot \boldsymbol{R}_j) \right|^2 \\ &+ (\pi/2) \left(\sum_{j=1}^{N} q_j \boldsymbol{R}_j \right)^2. \end{aligned} \tag{2.12}$$

For completeness, we give the result when the unit cell contains N dipoles $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_N$ which interact in pairs via a potential

$$\begin{aligned} \phi^{d}(\boldsymbol{R}_\alpha - \boldsymbol{R}_\beta, \boldsymbol{\mu}_\alpha, \boldsymbol{\mu}_\beta) = \{&\boldsymbol{\mu}_\alpha \cdot \boldsymbol{\mu}_\beta |\boldsymbol{R}_\alpha - \boldsymbol{R}_\beta|^{-2} \\ &- 2(\boldsymbol{\mu}_\alpha \cdot (\boldsymbol{R}_\alpha - \boldsymbol{R}_\beta))(\boldsymbol{\mu}_\beta \cdot (\boldsymbol{R}_\alpha - \boldsymbol{R}_\beta)) |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|^{-4}\} l^{-2}. \end{aligned} \tag{2.13}$$

The lattice sum for this array

$$U^{d}(N; D_0, 1) = \tfrac{1}{2} \sum_{l\boldsymbol{n} \in D_0}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} \phi^{d}(\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}, \boldsymbol{\mu}_j, \boldsymbol{\mu}_k) \tag{2.14}$$

may be readily calculated from eq. (2.12) and is,

$$\begin{aligned} U^{d}(N, D_0, 1) = &(2l^2)^{-1} \sum_{\boldsymbol{n}}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} \boldsymbol{\mu}_j \cdot \boldsymbol{\mu}_k \exp(-\alpha^2 |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^2) |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^{-2} \\ &- l^{-2} \sum_{\boldsymbol{n}}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} (\boldsymbol{\mu}_j \cdot (\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n})) \\ &\times (\boldsymbol{\mu}_k \cdot (\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n})) |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^{-4} \\ &\times \exp(-\alpha^2 |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^2)(1 + \alpha^2 |\boldsymbol{R}_j - \boldsymbol{R}_k + \boldsymbol{n}|^2) \\ &+ \pi l^{-2} \sum_{n \neq 0} |\boldsymbol{n}|^{-2} \exp(-\pi^2 \boldsymbol{n}^2 \alpha^{-2}) \left| \sum_{j=1}^{N} (\boldsymbol{n} \cdot \boldsymbol{\mu}_j) \exp(2\pi i \boldsymbol{n} \cdot \boldsymbol{R}_j) \right|^2 \\ &+ \tfrac{1}{2} l^{-2} \pi \left(\sum_{j=1}^{N} \boldsymbol{\mu}_j \right)^2 - \alpha^2 N \mu^2 l^{-2}, \end{aligned} \tag{2.15}$$

where $\mu = |\boldsymbol{\mu}_j|$ is the dipole moment. For applications in computer simulation, the value of α^2 is usually chosen so that only the largest term in the double sums over i, j contributes to the desired order of accuracy. Let δ be the maximum allowable error. Then α is chosen so that

$$E_l(\alpha^2/4) \ll \delta. \tag{2.16}$$

For the charge case, $E_l(z)$ has the asymptotic expansion

$$E_l(z) \simeq \mathrm{e}^{-z} \sum_{r=1}^{\infty} (-1)^{r-1} z^{-r} (r-1)!, \quad \text{as } |z| \to \infty,$$

so that α^2 should be chosen somewhat larger than the root of

$$\exp(-z/4) = \delta z/4. \tag{2.17}$$

3. Direct summation of Coulomb and dipolar lattice sums

In section 2, we have represented our lattice sums using the vector $\boldsymbol{r} = (x, y)$, $\boldsymbol{\mu} = (\mu_x, \mu_y)$. In two dimensions, we may with profit use the alternative representation of $\boldsymbol{r}$ as the complex variable $z = x + \mathrm{i}y$ and $\boldsymbol{\mu}$ as the complex variable $\mu = \mu_x + \mathrm{i}\mu_y$. The interaction between two charges q_α, q_β is then

$$\phi^{\mathrm{c}}_{\alpha\beta} = -q_\alpha q_\beta \,\mathrm{Re}\,\log(z_\alpha - z_\beta) \tag{3.1}$$

and between two dipoles

$$\phi^{\mathrm{d}}_{\alpha\beta} = -\mathrm{Re}\,\mu_\alpha \mu_\beta (z_\alpha - z_\beta)^{-2}. \tag{3.2}$$

In what follows, it will be convenient to define

$$\omega_1 = l/2, \tag{3.3a}$$

$$\omega_2 = \mathrm{i}l/2 = \mathrm{i}\omega_1. \tag{3.3b}$$

We consider the large, simply connected region D of the complex plane, with simple closed boundary curve Γ, the region D being tesselated into square cells of side l with centres at the points

$$\Omega_{m,n} = 2m\omega_1 + 2n\omega_2. \tag{3.4}$$

We consider those values of $\boldsymbol{n} = (m, n)$ for which $\Omega_{m,n}$ lies within D. We have, for the charge sum (2.1) over D

$$U^{\mathrm{c}}(N; D, 1) = -\tfrac{1}{2} \sum_{\Omega_{m,n} \in D}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} q_j q_k \,\mathrm{Re}\,\log(z_j - z_k - \Omega_{m,n}) \tag{3.5}$$

and for the dipole sum (2.14)

$$U^{\mathrm{d}}(N; D, 1) = -\tfrac{1}{2}\,\mathrm{Re} \sum_{\Omega_{m,n} \in D}{}' \sum_{j=1}^{N} \sum_{k=1}^{N} \mu_j \mu_k (z_j - z_k - \Omega_{m,n})^{-2}. \tag{3.6}$$

We first repeat our calculations over the circular region D_0, rewriting eq. (3.5) as

$$U^{c}(N; D_0, 1) = -\tfrac{1}{2}\,\mathrm{Re} \sum_{\substack{\Omega_{m,n}\in D_0\\ \Omega_{m,n}\neq 0}} \sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \{\log(z_j - z_k - \Omega_{m,n}) - \log(-\Omega_{m,n}) + (z_j - z_k)\Omega_{m,n}^{-1} + \tfrac{1}{2}(z_i - z_j)^2\Omega_{m,n}^{-2}\} - \tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k\neq j} q_j q_k \log(z_j - z_k). \tag{3.7}$$

This may be done because

$$\sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \log(-\Omega_{m,n}) = \sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \Omega_{m,n}^{-1}(z_j - z_k) = 0 \tag{3.8a}$$

by the charge neutrality condition and

$$\sum_{\Omega_{m,n}\in(D_0\cap 0)} (z_j - z_k)^2 \Omega_{m,n}^{-2} = 0, \tag{3.8b}$$

by symmetry when D_0 is a circle (or square). However, the terms in the sum (3.7) behave like $\Omega_{m,n}^{-3}$ as $|\Omega_{m,n}|\to\infty$ and so the series is an absolutely and uniformly convergent one for $z = z_j - z_k \neq \Omega_{m,n}$. We may thus take the radius of D_0 to infinity and derive the value of eq. (3.7) as

$$U^{c}(N; D_0, 1) = -\tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \log\{\sigma(z_j - z_k, \omega_1, \omega_2)/(z_j - z_k)\} - \tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k\neq j} q_j q_k \log(z_j - z_k), \tag{3.9}$$

where[11] $\sigma(z, \omega_1, \omega_2)$ is the pseudo-periodic sigma function of Weierstrass, with its two complex periods ω_1 and ω_2. Likewise the dipole sum (3.6) may be written as

$$U^{d}(N; D_0, 1) = -\tfrac{1}{2}\,\mathrm{Re}\sum_{\Omega_{m,n}\in D_0\cap 0}\sum_{j=1}^{N}\sum_{k=1}^{N} \mu_j\mu_k\{(z_j - z_k - \Omega_{m,n})^2 - \Omega_{m,n}^{-2}\} - \tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k\neq j}\mu_j\mu_k (z_j - z_k)^{-2} \tag{3.10}$$

$$= -\tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k=1}^{N}\mu_j\mu_k\{\wp(z_j - z_k, \omega_1, \omega_2) - (z_j - z_k)^{-2}\} - \tfrac{1}{2}\,\mathrm{Re}\sum_{j=1}^{N}\sum_{k\neq j}\mu_j\mu_k(z_j - z_k)^{-2}, \tag{3.11}$$

where $\wp(z, \omega_1, \omega_2)$ is the elliptic ρ-function of Weierstrass. As

$$\lim_{z\to 0}(\wp(z) - z^{-2}) = 0,$$

eq. (3.11) may be written as

$$U^{d}(N, D_0, 1) = -\mathrm{Re} \sum_{1\le j<k\le N} \mu_j\mu_k \wp(z_j - z_k, \omega_1, \omega_2). \tag{3.12}$$

Likewise, as

$$\lim_{z\to 0} z^{-1}\sigma(z) = 1,$$

eq. (3.9) may be written as

$$U^{c}(N; D_0, 1) = -\mathrm{Re} \sum_{1\le j<k\le N} q_j q_k \log\{\sigma(z_j - z_k)\}. \tag{3.13}$$

We noted earlier that the lattice sum for the Coulomb potential, eq. (2.12) is not translationally invariant, because of the presence of terms

$$-(\pi/2)q_j q_k(\boldsymbol{R}_j - \boldsymbol{R}_k)^2$$

for each pair of particles. It is interesting to see what happens to this term when $\boldsymbol{R}_j - \boldsymbol{R}_k$ is increased by an amount (1, 0). The change is

$$-\pi q_j q_k(X_j - X_k + \tfrac{1}{2}). \tag{3.14}$$

This corresponds to adding $2\omega_1$ to $z_j - z_k$ in eq. (3.13). The value of eq. (3.14) should equal

$$-\mathrm{Re}\{\log\{\sigma(z + 2\omega_1)\} - \log\{\sigma(z)\}\}.$$

However[11]),

$$\sigma(z + 2\omega_1) = -\exp(-2\eta_1(z + \omega_1))\sigma(z), \tag{3.15}$$

where

$$\eta_1 = \zeta(\omega_1)$$

and $\zeta(z)$ is the ζ-function of Weierstrass, defined as

$$\zeta(z) = \frac{\mathrm{d}}{\mathrm{d}z}(\log \sigma(z)) = z^{-1} + \sum_{\Omega_{m,n}\neq 0} ((z - \Omega_{m,n})^{-1} + \Omega_{m,n}^{-1} + z\Omega_{m,n}^{-2}).$$

In our case $\omega_2 = \mathrm{i}\omega_1 = \mathrm{i}l/2$, so that

$$\zeta(\omega_1) = \omega_1^{-1} + \sum_{\Omega_{m,n}\neq 0} \{(-(2m-1)\omega_1 - 2n\omega_2)^{-1} + (2m\omega_1 + 2n\omega_2)^{-1} + \omega_1(2m\omega_1 + 2n\omega_2)^{-2}\}$$

and

$$\begin{aligned}\zeta(\omega_2) &= (\mathrm{i}\omega_1)^{-1} + \sum_{\Omega_{m,n}\neq 0} \{(-2m\omega_1 - (2n-1)\omega_2)^{-1} + \mathrm{i}^{-1}(2n\omega_1 - 2m\mathrm{i}\omega_1)^{-1} \\ &\quad - \mathrm{i}\omega_1(2n\omega_1 - 2m\omega_2)^{-2}\} \\ &= -\mathrm{i}\zeta(\omega_1).\end{aligned}$$

However, since[11])

$$\omega_2\zeta(\omega_1) - \omega_1\zeta(\omega_2) = \mathrm{i}\pi/2,$$

we have

$$\zeta(\omega_1) = \pi/(2l).$$

Thus, from eq. (3.15),

$$\log\sigma(z + 2\omega_1) - \log\sigma(z) = \mathrm{Re}(-\pi l^{-1}(z + l/2)), \tag{3.16}$$

which is the same as (3.14), confirming the correctness of using convergence factors in section 2.

From a computational point of view, it is preferable to express the slowly convergent Weierstrass functions in terms of the rapidly Jacobi theta functions. We begin with the dipole sum (3.12). We have, in the notation of ref. 11,

$$\wp(z) = e_3 + \gamma^2/\mathrm{sn}^2(\gamma z), \tag{3.17}$$

with γ and e_3 to be determined and where sn is the normal Jacobi elliptic function. $\wp(z)$ has double poles at the points $\Omega_{m,n} = (2m + 2\mathrm{i}n)l/2$. $\mathrm{sn}(\gamma z)$ has zeros at the points

$$\gamma z = 2mK(k^2) + 2n\mathrm{i}K(1 - k^2),$$

where $K(k^2)$ is the complete elliptic integral. Evidently, $k^2 = \frac{1}{2}$ and

$$\gamma = 2K(\tfrac{1}{2})/l. \tag{3.18}$$

The 3 invariants e_1, e_2 and e_3 are then given by[12])

$$e_1 = \tfrac{1}{2}\gamma^2, \quad e_2 = 0, \quad e_3 = -\tfrac{1}{2}\gamma^2. \tag{3.19}$$

The sn function is conveniently computed in terms of the Jacobi theta functions. We have

$$\mathrm{sn}(\gamma Z) = 2^{\frac{1}{4}}\theta_1(\pi\gamma z/(2K(\tfrac{1}{2})), \mathrm{e}^{-\pi})/\theta_4(\pi\gamma z/(2K(\tfrac{1}{2})), \mathrm{e}^{-\pi}). \tag{3.20}$$

Thus

$$\wp(z) = -\tfrac{1}{2}\gamma^2 + 2^{-\frac{1}{2}}\gamma^2\theta_4^2(\pi z/l, \mathrm{e}^{-\pi})/\theta_1^2(\pi z/l, \mathrm{e}^{-\pi}). \tag{3.21}$$

The lattice sum for the Coulomb interaction may be similarly simplified. We have[11])

$$\sigma(z) = 2\omega_1/(\pi\theta_1') \exp(\eta_1 z^2/2\omega_1)\theta_1(\pi z/(2\omega_1), q).$$

For our case

$$q = \mathrm{e}^{-\pi}, \quad \eta_1 = \pi/(2l), \quad \omega_1 = l/2, \quad \theta_1' = 2(K(\tfrac{1}{2})/\pi)^{\frac{3}{2}},$$

so that

$$\sigma(z) = 2\pi^{-\frac{1}{2}} K(\tfrac{1}{2})^{\frac{3}{2}} \exp(\tfrac{1}{2}\pi z^2 l^{-2})\theta_1(\pi z/l, e^{-\pi}). \tag{3.22}$$

For completeness, we give the rapidly convergent Fourier series for the θ_1 and θ_4 functions. We have

$$\theta_4(z, q) = \sum_{n=-\infty}^{\infty} (-1)^n q^{n^2} e^{2niz} = 1 + 2\sum_{n=1}^{\infty} q^{n^2} \cos 2nz,$$

$$\theta_1(z, q) = -i\, e^{iz + i\pi\tau/4}\, \theta_4(z + \tfrac{1}{2}\pi\tau, q), \tag{3.23a}$$

where

$$\tau = -i\pi^{-1} \log q = i \quad \text{in our case.}$$

Thus

$$\theta_1(z, q) = 2\sum_{n=0}^{\infty} (-1)^n q^{(n+\frac{1}{2})^2} \sin(2n+1)z. \tag{3.23b}$$

Since, for our case, $\mathrm{Im}(z) > -\pi(-l < y_j - y_k < l)$, we can see how rapidly these series converge. If δ is the maximum allowable error, the series may be truncated when $n > N_0$, where N_0 is the greatest integer less than

$$1 + (1 - \pi^{-1} \log \delta)^{\frac{1}{2}}. \tag{3.24}$$

If δ is 10^{-8}, N_0 is 3. To underline this point, for our purposes

$$\theta_1(z, q) = 1 - q(e^{2iz} + e^{-2iz}) + q^4(e^{4iz} + e^{-4iz}) - q^9(e^{6iz} + e^{-6iz}), \tag{3.25}$$

since $q = e^{-\pi}$. Eq. (3.25) is amenable to very rapid computation.

Let F_x and F_y be the components of the force on charge q_j due to charge q_k. Using the Cauchy–Riemann equations, we have

$$F_x - iF_y = q_j q_k \zeta(z_j - z_k). \tag{3.26}$$

Using eq. (3.22), this becomes

$$F_x - iF_y = q_j q_k \left\{ \pi(z_j - z_k) l^{-2} + \frac{\pi l^{-1} \theta_1'(\pi(z_j - z_k)/l, e^{-\pi})}{\theta_1(\pi(z_j - z_k)/l, e^{-\pi})} \right\}. \tag{3.27}$$

Thus in implementing a molecular dynamics study of two-dimensional Coulomb systems, it may well be advantageous to use complex variables in view of the rapidly convergent series (3.25) for θ_1, and its derivative θ_1'.

We now extend these results to the case when the sum is over an arbitrary, simply connected region whose boundary is the simple, closed curve Γ. We study the Coulomb case and write

$$\begin{aligned} U^{c}(N; D, 1) = -\tfrac{1}{2}\,\mathrm{Re} \sum_{\Omega_{m,n}\in D_0\sim D} \sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \{\log(z_j - z_k - \Omega_{m,n}) \\ - \log(-\Omega_{m,n}) + \Omega_{m,n}^{-1}(z_j - z_k) + \tfrac{1}{2}(z_j - z_k)^2 \Omega_{m,n}^{-2}\} \\ -\mathrm{Re} \sum_{1\le j<k\le N} q_j q_k \log(z_j - z_k) \\ +\tfrac{1}{4}\,\mathrm{Re} \sum_{\Omega_{m,n}\in D_0\sim D} \sum_{j=1}^{N}\sum_{k=1}^{N} q_j q_k \Omega_{m,n}^{-2}(z_j - z_k)^2, \end{aligned} \tag{3.28}$$

Here D_0 is a circle within D, such that the ratio of the two areas is finite and greater than zero. Let Γ_0 be the boundary curve of D_0. We define

$$z_0^* = \max_{z\in\Gamma_0} |z| \quad \text{and} \quad z^* = \max_{z\in\Gamma} |z|.$$

It suffices to consider

$$S = \sum_{\Omega_{m,n}\in D_0\sim D} (ml + \mathrm{i}nl)^{-2}. \tag{3.29}$$

Let M be an integer such that

$$Ml \le |z_0^*| \le (M+1)l.$$

Then

$$S = l^{-2}M^{-2} \sum_{\Omega_{m,n}\in D_0\sim D} (m/M + \mathrm{i}n/M)^{-2}.$$

As $|z_0^*|\to\infty$, $M^{-1}\to 0$ and $(m/M,\, n/M)$ tend to continuous variables (x, y). Thus

$$\lim_{|z_0^*|\to\infty} S = l^{-2} \int_{D'}\int_{-D_0'} \mathrm{d}x\,\mathrm{d}y (x+\mathrm{i}y)^{-2}, \tag{3.30}$$

where D_0' is the unit circle and D' the region D suitably scaled. We consider the integral over the cut region in fig. 1, which tends to S as the width of the cut tends to zero. Then

$$\begin{aligned} S &= -\tfrac{1}{2}l^{-2} \int\int_{D^*} \mathrm{d}x\,\mathrm{d}y \left\{\frac{\partial}{\partial x}(x+\mathrm{i}y)^{-1} - \mathrm{i}\frac{\partial}{\partial y}(x+\mathrm{i}y)^{-1}\right\} \\ &= -\tfrac{1}{2}l^{-2} \int_{\Gamma^*} (\mathrm{i}\,\mathrm{d}x + \mathrm{d}y)(x+\mathrm{i}y)^{-1}, \end{aligned} \tag{3.31}$$

by Green's theorem, where Γ^* is the boundary of D^*. The integrals along the

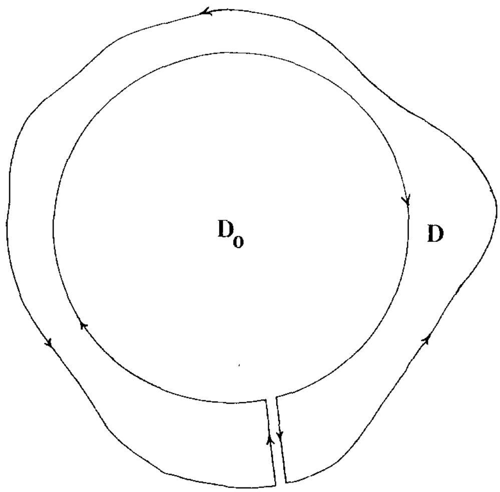

Fig. 1. Showing the region of integration for eq. (3.31).

cut cancels and the integral along the circular arc is easily calculated to be zero. We thus have

$$S = -\tfrac{1}{2}\mathrm{i}l^{-2} \int_{\Gamma^*} (\mathrm{d}x - \mathrm{i}\,\mathrm{d}y)(x+\mathrm{i}y)^{-1} = -\tfrac{1}{2}\mathrm{i}l^{-2} \int_{\Gamma^*} z^{-1}\,\overline{\mathrm{d}z}. \tag{3.32}$$

Thus we obtain the general result that

$$U^{\mathrm{c}}(N; D, 1) = U^{\mathrm{c}}(N; D_0, 1) + \mathrm{Re}\Big\{\mathrm{i}/(4l^2)\Big(\sum_{j=1}^{N} q_j z_j)^2 \int_{\Gamma} z^{-1}\,\overline{\mathrm{d}z}\Big\}. \tag{3.33}$$

The corresponding result for dipoles is then

$$U^{\mathrm{d}}(N; D, 1) = U^{\mathrm{d}}(N; D_0, 1) + \mathrm{Re}\Big\{\mathrm{i}/(4l^2)\Big(\sum_{j=1}^{N} \mu_j\Big)^2 \int_{\Gamma} z^{-1}\,\overline{\mathrm{d}z}\Big\}. \tag{3.34}$$

As an example, let D be the rectangle $-a < x < a$, $-b < y < b$. Keeping careful track of the branches of the logarithm

$$\int_{\Gamma} z^{-1}\,\overline{\mathrm{d}z} = 2\pi\mathrm{i} - 8\mathrm{i}\tan^{-1}(b/a). \tag{3.35}$$

We note that if $b = a$, the rectangle becomes a square, and

$$U^{\mathrm{c}}(N; D, 1) = U^{\mathrm{c}}(N; D_0, 1), \quad U^{\mathrm{d}}(N; D, 1) = U^{\mathrm{d}}(N; D_0, 1),$$

as they should.

4. Dielectric properties of two-dimensional dipolar systems

In the previous section, we have seen that the Hamiltonian of a two-dimensional periodic Coulomb system depends on the shape of the system's boundary and computed that variation explicitly. We shall see here that it also depends on the system's surroundings. We begin with the derivation of the polarization fluctuation formula. The mean polarization of the central cell, $\boldsymbol{P}$, caused by an infinitesimal, externally applied constant electric field $\boldsymbol{E}$ is

$$\boldsymbol{P} = (\rho/k_{\mathrm{B}}T)\boldsymbol{E}\cdot\{\langle \boldsymbol{M}\boldsymbol{M}\rangle - \langle \boldsymbol{M}\rangle\langle \boldsymbol{M}\rangle\}N^{-1}, \tag{4.1}$$

where

$$\boldsymbol{M} = \sum_{j=1}^{N} \boldsymbol{\mu}_j$$

is the cell dipole moment. The symbol $\langle\ \ \rangle$ represents an ensemble average. Now imagine that the periodic region, a circle of radius a, is surrounded by a much larger circular annulus of outer radius b, consisting of uniform dielectric material of dielectric constant ϵ'. In this case, the instantaneous uniform polarization $\boldsymbol{M}/l^2$ will polarize the outer dielectric, producing a reaction field which will interact with $\boldsymbol{M}$, thus altering the Hamiltonian.

From the point of view ɩf computing $\boldsymbol{P}$, we can also regard the inner circular region as a uniform dielectric of dielectric constant ϵ, whose polarization we wish to calculate. If we now place the composite region in a uniform, infinitesimal electric field $\boldsymbol{E}_0$ and compute $\boldsymbol{P}$ by eq. (4.1) and by macroscopic electrostatics, a comparison of the two formulae will give us a formula for ϵ. We first solve the potential theory problem, finding for the field $\boldsymbol{E}$ within the inner circular region,

$$\boldsymbol{E} = 4\epsilon'(\epsilon'+1)^{-1}(\epsilon'+\epsilon)^{-1}\boldsymbol{E}_0 - 2\pi(\epsilon+\epsilon')^{-1}\boldsymbol{M}l^{-2}. \tag{4.2}$$

To find the reaction field $\boldsymbol{E}_R$, we set $\boldsymbol{E}_0 = 0$, $\epsilon = 1$ in eq. (4.2) and obtain

$$\boldsymbol{E}_{\mathrm{R}} = -2\pi(1+\epsilon')^{-1}\boldsymbol{M}l^{-2}, \tag{4.3}$$

which interacts with $\boldsymbol{M}$ with an energy $-\frac{1}{2}\boldsymbol{M}\cdot\boldsymbol{E}_R$. Thus

$$\begin{aligned} U^{\mathrm{d}}(N;D_0,\epsilon') - U^{\mathrm{d}}(N;D_0,1) &= \pi\boldsymbol{M}^2l^{-2}\{(\epsilon'+1)^{-1}-\tfrac{1}{2}\} \\ &= -\tfrac{1}{2}\pi\boldsymbol{M}^2(\epsilon'-1)(\epsilon'+1)^{-1}l^{-2}. \end{aligned} \tag{4.4}$$

To compute the field $\boldsymbol{E}$ which interacts with the dipoles when the field $\boldsymbol{E}_0$ is applied, we set $\epsilon = 1$, $\boldsymbol{M} = 0$ in eq. (4.2) to obtain

$$\boldsymbol{E} = 4\epsilon'(\epsilon'+1)^{-2}\boldsymbol{E}_0, \tag{4.5}$$

which is the field to be inserted into eq. (4.1).

To compute $\boldsymbol{P}$ by macroscopic electrostatics, we set $\boldsymbol{M} = 0$ in eq. (4.2) and compute the local field

$$\pi \boldsymbol{E}' = 4\epsilon'(\epsilon' + 1)^{-1}(\epsilon' + \epsilon)^{-1}\boldsymbol{E}_0, \tag{4.6}$$

from which $\boldsymbol{P}$ is computed as

$$\boldsymbol{P} = (\epsilon - 1)\boldsymbol{E}'/2\pi = 2\epsilon'(\epsilon - 1)\boldsymbol{E}_0(\epsilon' + 1)^{-1}(\epsilon + \epsilon')^{-1} \tag{4.7}$$

$$= 2\epsilon'(\epsilon - 1)(\epsilon' + 1)^{-1}(\epsilon + \epsilon')^{-1}\boldsymbol{E}_0 \cdot \mathbf{1} \tag{4.8}$$

where the (assumed) isotropy of the material is introduced via the unit tensor $\mathbf{1}$. Comparing the co-efficients of $\boldsymbol{E}_0$ in eq. (4.8) and the combination of (4.5) and (4.1) and taking the trace of the tensors, we obtain the fluctuation formula

$$\frac{(\epsilon - 1(\epsilon' + 1)}{2(\epsilon + \epsilon')} = \pi\rho\mu^2/(2k_{\mathrm{B}}T)[\{\langle \boldsymbol{M}^2\rangle - \langle \boldsymbol{M}\rangle^2\}/N\mu^2]. \tag{4.9}$$

The contents of the square bracket is the two-dimensional analogue of the g-factor of Kirkwood, which we write as $g(\epsilon')$ to remind us that it depends on ϵ' through the dependence of the Hamiltonian (4.4) on ϵ'. Setting $\epsilon' = \epsilon$ in eq. (4.9), the two-dimensional analogue of Kirkwood's formula is obtained, while setting $\epsilon' = 1$ leads to the Clausius–Mosotti equation, and $\epsilon' = \infty$ the formula of Debye.

Using the methods of ref. 8, it may then be shown that the value of ϵ is independent of ϵ', although we see no point in repeating that analysis.

References

1) E.R. Smith and J.W. Perram, Molec. Phys. **39** (1975) 31.
2) S.W. de Leeuw, J.W. Perram and E.R. Smith, Proc. Roy. Soc. **A373** (1980) 27.
3) B.U. Felderhof, Physica **101A** (1981) 275.
4) D.M. Heyes, J. Chem. Phys. **74** (1981) 1924.
5) E.R. Smith, Proc. Roy. Soc. **A** (in press).
6) S.W. de Leeuw and J.W. Perram, Physica **107A** (1981) 179.
7) E.L. Pollock and B.J. Alder, Physica **102A** (1980) 1.
8) S.W. de Leeuw, J.W. Perram and E.R. Smith, Proc. Roy. Soc. **A373** (1980) 57.
9) G. Bossis, B. Quentrec and C. Brot, Molec. Phys. **39** (1980) 1233.
10) Y. Saito and H. Müller-Krumbhaar, Phys. Rev. **B23** (1981) 308.
11) E.T. Whittaker and G.N. Watson, A Course of Modern Analysis (Cambridge Univ. Press, London, 1978) pp. 429–490.
12) P.F. Byrd and M.D. Friedman, A Handbook of Elliptic Integrals for Engineers and Physicists (Springer, Berlin, 1971) pp. 308–310.